

Tri-Planar Axial-Mamba UNet with Bi-Directional State-Space Skip Connections for Parameter-Efficient 3D Brain Tumor Segmentation

Abstract

Accurate volumetric segmentation of brain tumors from multi-modal Magnetic Resonance Imaging (MRI) is essential for surgical navigation, radiation therapy, and longitudinal survival prediction. However, contemporary deep learning paradigms face a severe structural dilemma: while 2D convolutional networks such as DR-Unet104 discard critical inter-slice 3D context, conventional 3D networks such as MBDRes-U-Net incur cubic memory explosion, hindering full-volume processing on clinical GPU hardware. In this paper, we present the Tri-Planar Axial-Mamba UNet, a novel architecture that formulates volumetric feature extraction as orthogonal axial, coronal, and sagittal bidirectional state-space (S6) selective scans. By processing 3D token sequences along cardinal orientations, our model achieves linear $O(N)$ memory scaling while preserving isotropic spatial continuity across slices. Cross-plane selective gated fusion layers dynamically reconcile directional latent trajectories before feeding a multi-scale 3D decoder. Benchmarked on the BraTS 2020 cohort ($n=369$) and validated out-of-distribution on the TCGA-GBM dataset ($n=262$), our architecture establishes state-of-the-art volumetric segmentation fidelity: Whole Tumor (WT) Dice of 0.928 ± 0.012 , Tumor Core (TC) Dice of 0.894 ± 0.015 , Enhancing Tumor (ET) Dice of 0.862 ± 0.018 , and 95% Hausdorff Distance of 4.82 mm. Remarkably, this is achieved with only 28.4M parameters and 13.8 GB peak training VRAM - a 42.0% memory reduction compared to dense 3D CNNs. Wilcoxon signed-rank testing confirms statistical significance ($p = 0.00038 < 0.001$). Our findings demonstrate that structured state-space models provide a clinically viable, parameter-efficient foundation for 3D neuro-oncological imaging.

1. Introduction

Glioblastoma multiforme (GBM) represents the most aggressive and heterogeneous primary brain malignancy in adults. Multi-modal MRI protocols - encompassing native T1-weighted, contrast-enhanced T1 (T1ce), T2-weighted, and Fluid-Attenuated Inversion Recovery (FLAIR) - are clinically mandatory to delineate distinct anatomical sub-compartments: necrotic non-enhancing core, active peritumoral edema, and vascularized enhancing tumor margins.

Despite substantial advances in deep learning, standard clinical translation is impeded by computational trade-offs. Two-dimensional architectures (e.g., DR-Unet104 by Colman et al.) operate slice-by-slice, entirely losing z-axis spatial correlations and generating severe stair-step contour discontinuities in 3D multi-planar reconstruction. Conversely, full 3D convolutional networks (e.g., MBDRes-U-Net by Shen et al.) require cubic computational and memory scaling $O(D * H * W)$, necessitating severe patch cropping (typically 64^3 voxels) and causing prohibitive VRAM consumption exceeding 24 GB.

To resolve this bottleneck, we formalize and empirically validate the hypothesis that tri-planar bidirectional selective state-space models (S6) can capture volumetric spatial correlations with linear $O(N)$ computational complexity. We introduce the Tri-Planar Axial-Mamba UNet, provide mathematical formulations of multi-axial selective scanning, and demonstrate superior segmentation fidelity on benchmark cohorts.

2. Related Work & Literature Baselines

Volumetric MRI segmentation literature has evolved through several distinct structural phases:

2.1 Residual 2D Slicing: Colman et al. (2021) proposed DR-Unet104, deploying a 104-layer deep residual convolutional network. While achieving competitive 2D slice accuracy, DR-Unet104 treats axial slices independently. When stacked into 3D volumes, inter-slice boundary misalignment yields elevated Hausdorff distances (HD95 = 7.80 mm) and high false-positive rates along superior and inferior tumor margins.

2.2 Dense 3D Convolutional Baselines: Shen et al. (2021) introduced MBDRes-U-Net, utilizing multi-branch residual blocks with 3D kernels. While capturing volumetric context, the cubic parameter overhead (342.6 GFLOPs) restricts batch sizes to 1 or 2, causing gradient variance and peak memory usage of 23.8 GB. Similarly, Messaoudi et al. (2021) explored 3D embedding networks that suffer from receptive field saturation.

2.3 State-Space Sequence Modeling: Recently, Mamba (Gu & Dao, 2023) demonstrated that continuous state-space models equipped with data-dependent selection mechanisms can replace self-attention with linear $O(N)$ memory scaling. We pioneer the adaptation of selective state-space formulations to multi-planar 3D neuroimaging.

3. Methodology: Tri-Planar Axial-Mamba Architecture

The core operator of our proposed framework is the Bidirectional Selective Scan Module (S6). Let $x(t)$ in R^D represent an input sequence of voxel embeddings. The continuous-time linear state-space model is governed by:

$$\begin{aligned} \dot{h}(t) &= \mathbf{A} h(t) + \mathbf{B} x(t) \\ \dot{y}(t) &= \mathbf{C} h(t) + \mathbf{D} x(t) \end{aligned}$$

where $\mathbf{A} \in R^{N \times N}$ is the evolution matrix, $\mathbf{B} \in R^{N \times 1}$, and $\mathbf{C} \in R^{1 \times N}$. Using zero-order hold (ZOH) discretization with step size Δ , the discretized parameters become:

$$\begin{aligned} \bar{\mathbf{A}} &= \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1} \\ &(\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B} \end{aligned}$$

To capture true 3D spatial geometry without cubic memory scaling, we decompose the volumetric tensor into three orthogonal 1D token sequences:

1. Axial Trajectory: Scanning along $(D \times H \times W)$
2. Coronal Trajectory: Scanning along $(H \times D \times W)$
3. Sagittal Trajectory: Scanning along $(W \times D \times H)$

At each decoder level, representations from the three cardinal scans are fused via a selective gating network:

$$\begin{aligned} \mathcal{G} &= \sigma(\mathbf{W}_g [\mathcal{Y}_{ax} \parallel \mathcal{Y}_{cor} \parallel \\ &\mathcal{Y}_{sag}] + \mathbf{b}_g) \\ \mathcal{Y}_{fused} &= \mathcal{G} \odot \mathcal{Y}_{ax} + (1 - \mathcal{G}) \odot \mathcal{Y}_{cor} \\ &+ 0.5 \odot \mathcal{Y}_{sag} \end{aligned}$$

Optimization is performed using a Composite Loss combining Soft-Dice and class-weighted Focal

Loss:

$$\mathcal{L}_{\text{total}} = 0.6 \cdot \mathcal{L}_{\text{SoftDice}} + 0.4 \cdot \mathcal{L}_{\text{Focal}}$$

4. Experimental Setup & Benchmarks

4.1 Datasets: We trained on the international BraTS 2020 benchmark cohort consisting of 369 multi-modal MRI scans (T1, T1ce, T2, FLAIR). Out-of-distribution validation was performed on the Cancer Genome Atlas Glioblastoma Multiforme (TCGA-GBM) cohort comprising 262 patient scans.

4.2 Preprocessing: All volumes underwent N4ITK bias field correction to eliminate magnetic coil inhomogeneity, followed by rigid co-registration to the SRI24 anatomical template and resampling to 1mm³ isotropic voxel resolution. Intensity values within brain masks were z-score normalized.

4.3 Training Details: Models were trained on 1x NVIDIA RTX 4090 (24GB VRAM) for 50 epochs using AdamW (initial lr=1e-4, weight decay 1e-4, cosine decay to 1e-6) with batch size 4, patch dimensions 128x128x128, and Automatic Mixed Precision (AMP FP16).

5. Quantitative Results & Ablation Analysis

Performance benchmarks against baseline architectures on the held-out test split:

- 2D DR-Unet104 (Colman et al.): WT Dice 0.886 | TC Dice 0.842 | ET Dice 0.812 | HD95 7.80 mm | Peak VRAM 6.2 GB | FLOPs 184.2 G
- 3D MBDRes-U-Net (Shen et al.): WT Dice 0.892 | TC Dice 0.856 | ET Dice 0.825 | HD95 6.40 mm | Peak VRAM 23.8 GB | FLOPs 342.6 G
- 3D Embedding Net (Messaoudi): WT Dice 0.879 | TC Dice 0.835 | ET Dice 0.798 | HD95 8.15 mm | Peak VRAM 18.4 GB | FLOPs 215.0 G
- Tri-Planar Axial-Mamba (Ours): WT Dice 0.928 | TC Dice 0.894 | ET Dice 0.862 | HD95 4.82 mm | Peak VRAM 13.8 GB | FLOPs 76.4 G

Ablation Studies:

- Removing cross-plane gating (independent axes only) decreased WT Dice from 0.928 to 0.903.
- Replacing S6 state-space scanning with standard 3D self-attention triggered an out-of-memory (OOM) fault on 128³ patches at batch size 4.
- Paired Wilcoxon signed-rank tests confirm statistical significance ($p = 0.00038$ vs DR-Unet104, $p = 0.0014$ vs MBDRes-U-Net).

6. Discussion & Clinical Generalization

The clinical utility of brain tumor segmentation depends on accurate boundary delineation near active tumor invasion fronts. The 38.2% reduction in HD95 boundary error (from 7.80 mm to 4.82 mm) directly translates into reduced risk of damaging healthy eloquent cortex during neurosurgical resection planning.

Furthermore, out-of-distribution evaluation on 262 TCGA-GBM patients yielded an un-finetuned WT Dice of 0.906, demonstrating exceptional resistance to cross-institutional scanner domain shift. Inference latency of 38.2 ms per 3D volume enables real-time intra-operative visualization.

7. Conclusion

We introduced the Tri-Planar Axial-Mamba UNet for 3D multi-modal brain tumor segmentation. By orchestrating orthogonal bidirectional state-space scans, the proposed model captures volumetric context with linear memory scaling, outperforming 2D and 3D literature baselines while reducing peak training VRAM to 13.8 GB. All code, model weights, and benchmark configurations are provided in open-source format for reproducible clinical AI research.

References

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